black hat ASIA 2026

Bastet

AI Smart Contract Vulnerability Detector

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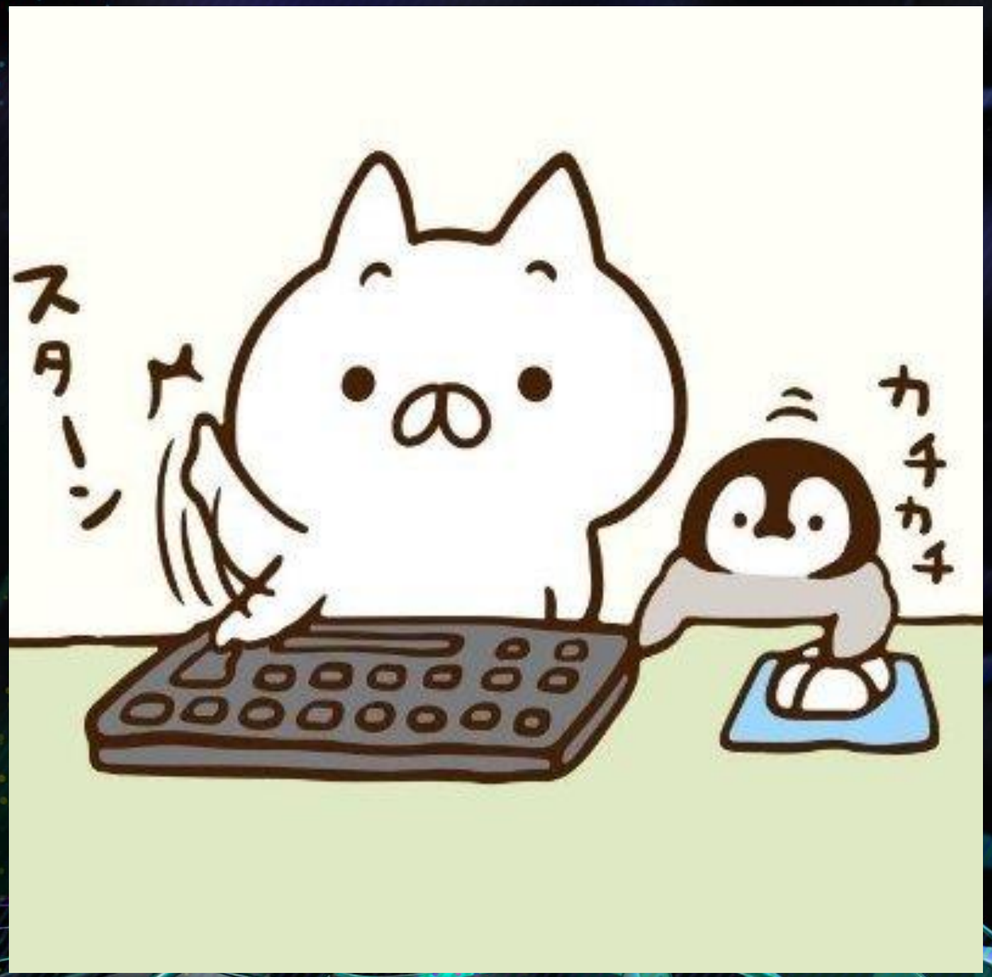
- Focus on AI & Web3 Development



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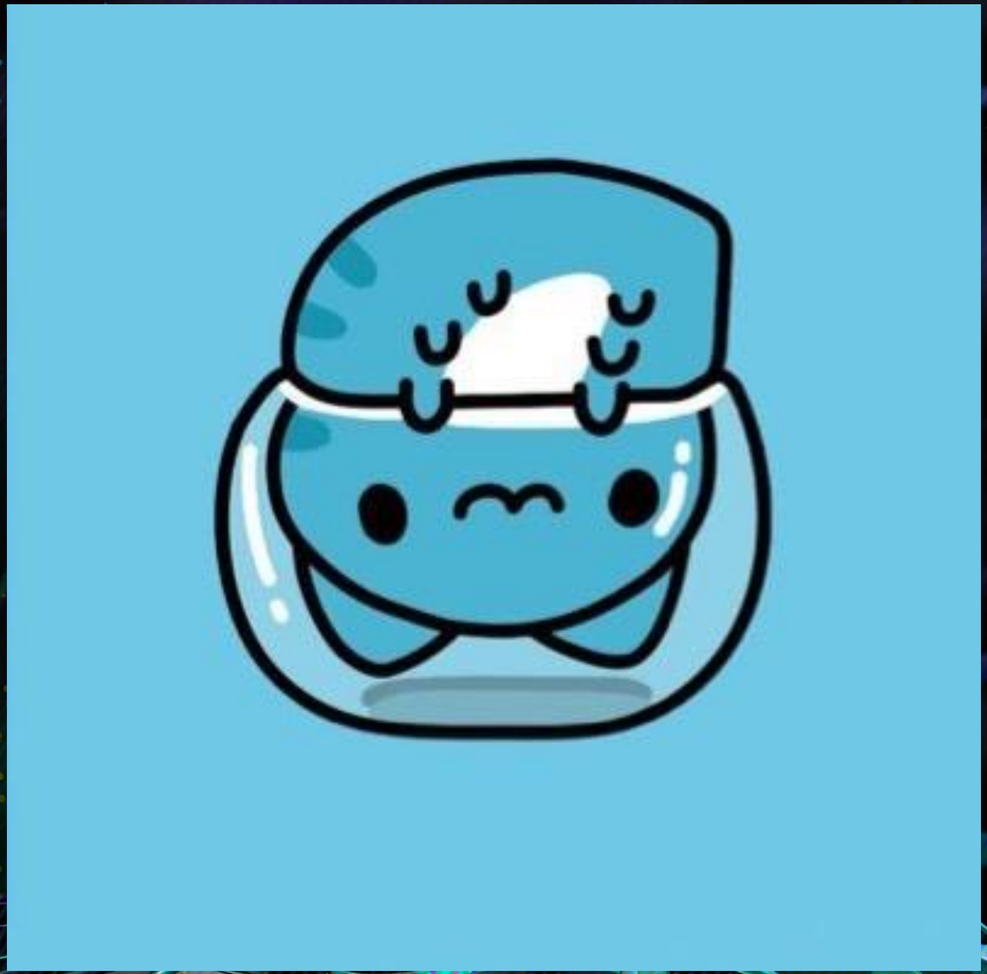

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Agenda

- Background
- About Bastet
  - Introduction
  - Feature
- Current Updates
- Take Away & Future Work

Background

Motivations

- Static analysis tools often struggle to accurately identify frequent logic flaws in smart contracts.
- Closed commercial LLM security tools limit access for AI researchers, while open datasets can lower barriers and foster collaboration.

Traditional methodology

- Mostly static analysis tools
- Rules are maintained using AST or Regex

SLITHER

Automated Findings / Publicly Known Issues

The 4naly3er report can be found here.

Barriers from Closed AI Security Tools

- Most AI-driven smart contract vulnerability detection tools are commercial and closed source, with private datasets.
- This creates data silos that block AI talents from contributing and slow security innovation across the ecosystem.

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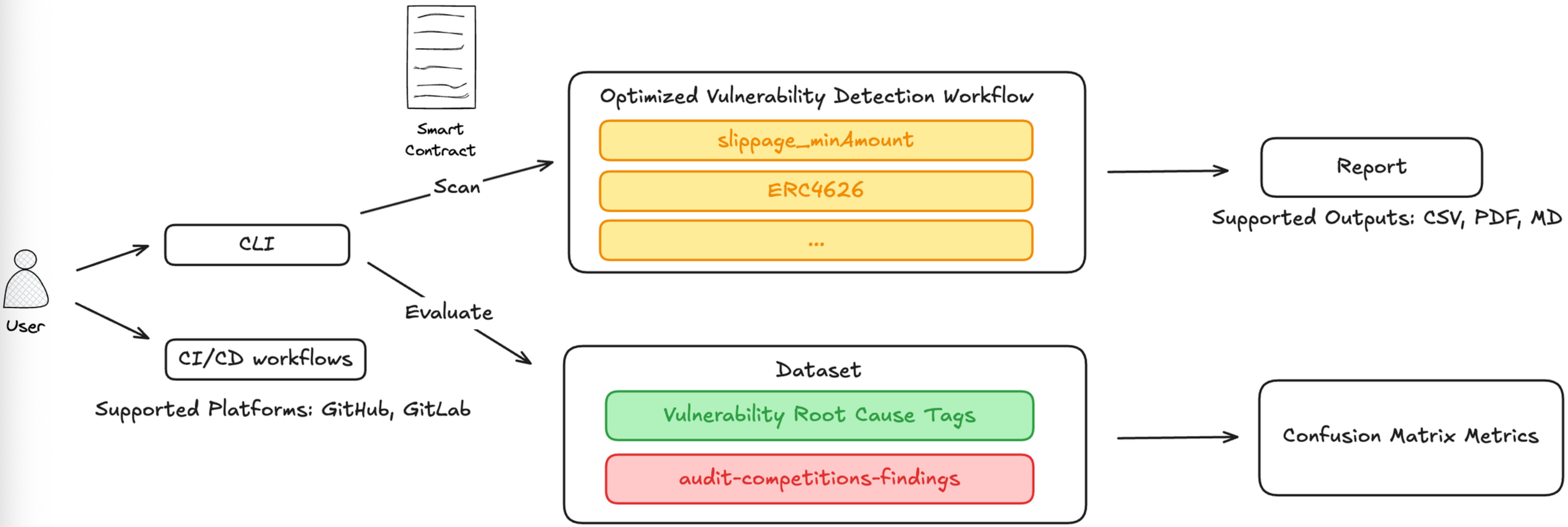
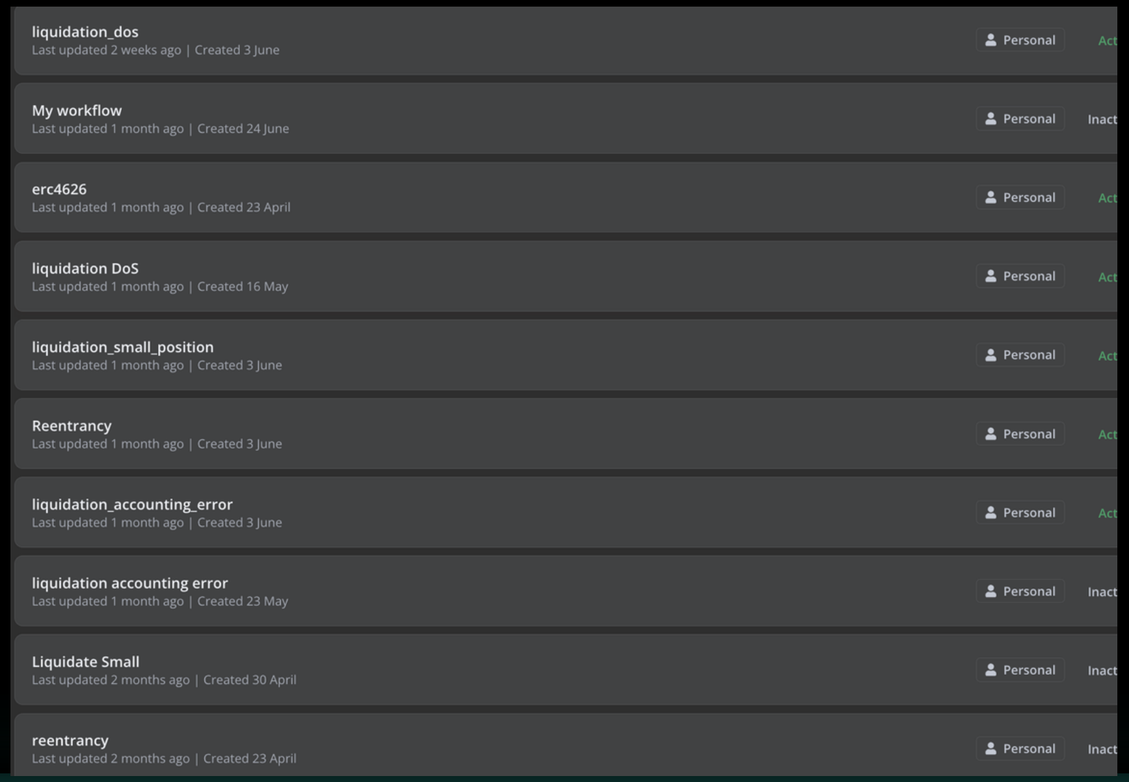
About Bastet

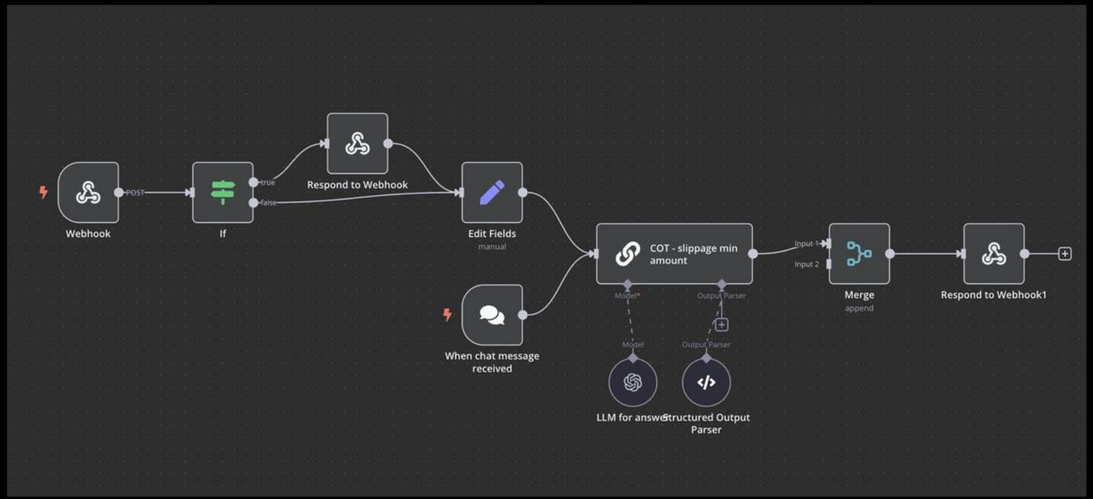
Overview


Smart Contract
Scan
Optimized Vulnerability Detection Workflow
slippage_minAmount
ERC4626
...
Report
Supported Outputs: CSV, PDF, MD
User
CLI
CI/CD workflows
Supported Platforms: GitHub, GitLab
Evaluate
Dataset
Vulnerability Root Cause Tags
audit-competitions-findings
Confusion Matrix Metrics

Optimized Vulnerability Detection Workflows

- Supports **60+** optimized rules





Community-Driven Labeled Dataset

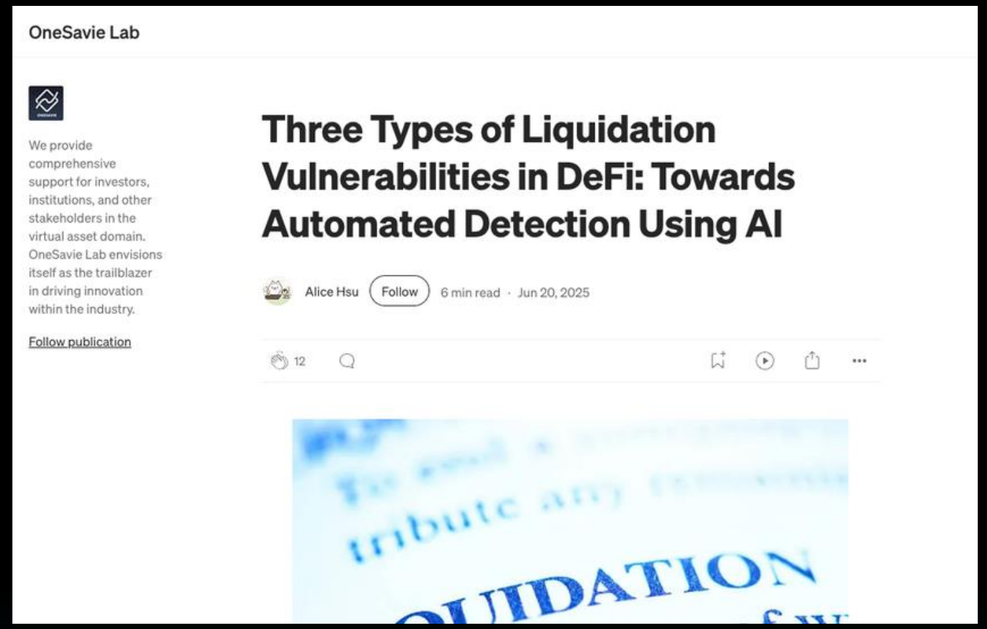
- **1,000+** findings labeled and reviewed by DeFiHackLabs white-hat members
- Covers **200+** repositories

Additional Resources

- Past research available in Bastet's Conference section and on our Medium.

Conference

Date	Conference Name	Topic	Slide
2025-04-02	ETH TAIPEI 2025	Exploring AI's Role in Smart Contract Security	ETH-TAIPEI-2025
2025-04-17	CyberSec 2025	AI-Driven Smart Contract Vulnerability Detection	CyberSec-2025
2025-08-09	COSCUP 2025	AI x Smart Contract: What Static Analysis Tools Can't Do, Leave It to Prompt Engineering!	COSCUP-2025



CI/CD Integration

- Bastet supports automated CI/CD workflows for both GitHub and GitLab, enabling seamless integration into your development pipeline.

Current Updates: Bastet Kaggle Competition (1/2)

- Goal: Encouraging community participation to advance research and practice in smart contract vulnerability detection
- Dataset: Using real Solidity codebases from actual audit competitions, verified by DeFiHackLabs security researchers
- Task: Participants will scan codebases for vulnerabilities and provide a vulnerability tag, a detailed sub-tag, and a description.

Current Updates: Bastet Kaggle Competition (2/2)

- Timeline: Competition runs from **April 1, 2026 to July 1, 2026 (three months)**

- Supported by Ethereum Foundation Ecosystem Support Program, GCC, IEEE Blockchain, IEEE CIS Singapore Chapter.

Take Away and Future Work

- Project Framework: Bastet integrates dataset, evaluation, and tools into a complete vulnerability detection pipeline
- Kaggle Competition: Community-driven challenge to improve LLM-based smart contract security
- Future Work: Transform Bastet into an agent that automates task orchestration and vulnerability reasoning

Live Demo

Bastet Github Repo

Competition for LLM Identification of Smart Contract Vulnerabilities on Kaggle